

**CHOOSING ROBUST LEADERSHIP: ENCOMPASSING THE BEST-OF-N
MODEL AND SWARM INTELLIGENCE OPTIMIZATION FOR
HETEROGENEOUS MULTIPLE AUTONOMOUS UNMANNED AERIAL
VEHICLE SYSTEMS**

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PREVIEW

LIST OF ABBREVIATIONS

ABC	Artificial Bee Colony Algorithm
ACO	Ant Colony Optimization
AGTO	Artificial Gorilla Troops Optimizer
AHA	Artificial Hummingbird Algorithm
ALO	Ant Lion Optimizer
ALSO	Artificial Lizard Search Optimization
AO	Aquila Optimizer
ASFA	Improved Artificial Fish Swarm Algorithm
AVOA	African Vultures Optimization Algorithm
BA	Bat-Inspired Algorithm
BCPA	Bee Collecting Pollen Algorithm
BMO	Bird Mating Optimizer
CHOA	Chimp Optimization Algorithm
CSA	Chameleon Swarm Algorithm
CSO	Cat Swarm Optimization
CS	Cuckoo Search
DA	Dragonfly Algorithm
DE	Dolphin Echolocation

DHO	Deer Hunting Optimization Algorithm
DMO	Dwarf Mongoose Optimization
DPO	Dolphin Partner Optimization
EEL	Eel Swarm Intelligence Algorithm
EHO	Elephant Herding Optimization
EPO	Emperor Penguin Optimizer
FA	Firefly Algorithm
FFOA	Fruit Fly Optimization Algorithm
FPA	Flower Pollination Algorithm
GEO	Golden Eagle Optimizer
GOA	Grasshopper Optimisation Algorithm
GSO	Glowworm Swarm Optimization
GWO	Grey Wolf Optimizer
HBA	Honey Badger Algorithm
HHO	Harris Hawks Optimization
HOA	Horse Herd Optimization Algorithm
HS	Hunting Search
IWPA	Improved Wolf Pack Algorithm
JS	Jellyfish Swarm Optimization
KH	Krill Herd
MBO-1	Migrating Birds Optimization
MBO	Marriage in Honey Bees Optimization
MB	Monarch Butterfly Optimization

MFA	Moth Flame Optimization
MPA	Marine Predators Algorithm
MS	Monkey Search Algorithm
PSO	Particle Swarm Optimization Algorithm
QANA	Quantum-based Avian Navigation Optimizer Algorithm
RFO	Red Fox Optimization Algorithm
ROA	Raccoon Optimization Algorithm
RSA	Reptile Search Algorithm
RSO	Rat Swarm Optimizer
SFLA	Shuffled Frog Leaping Algorithm
SHO	Spotted Hyena Optimizer
SMA	Slime Mould Algorithm
SMO	Starling Murmuration Optimizer
SSA-1	Salp Swarm Algorithm
SSA	Squirrel Search Algorithm
TCO	Termite Colony Optimization
TSA	Tunicate Swarm Algorithm
TSOA	Tree Search Optimization Algorithm
VOA	Virus Optimization Algorithm
WOA	Whale Optimization Algorithm
WSO	Wasp Swarm Algorithm
ZOA	Zebra Optimization Algorithm

ABSTRACT

Warnakulasooriya, Kudamuhandiramlage Harith Kolitha, Ph.D., University of South Alabama, May 2025. Choosing Robust Leadership: Encompassing the Best-of-N Model and Swarm Intelligence Optimization for Heterogeneous Multiple Autonomous Unmanned Aerial Vehicle Systems. Chair of Committee: Aviv Segev, Ph.D.

In recent years, numerous coordinated unmanned aerial vehicle systems have been extensively studied and developed for a variety of applications. Researchers aimed to expand the size of the collective group by incorporating diverse agents. However, due to connectivity constraints, a hierarchical control structure has been proposed for the autonomous operation of large unmanned aerial vehicle systems. A significant limitation of this hierarchical control structure is leader failure, which may lead to cascading failures within the entire group. The primary objective of this dissertation is to introduce a distributed, fully autonomous, and efficient leader selection algorithm supported by swarm intelligence. The proposed methodology is formalized within the framework of the best-of-n model. This dissertation comprises four key sections. Firstly, an in-depth analysis of swarm intelligence and behavior has been conducted. Building upon this analysis, a novel swarm intelligence algorithm, termed "Tree Search Optimization Algorithm (TSOA)," is proposed. Utilizing the TSOA, a new leader selection algorithm is introduced, along with an automatic method for capturing fallen UAVs, which serves as the triggering event for the algorithm. Concurrently, a task allocation algorithm is put forth for large UAV systems based on the C-Net Protocol.

The swarm intelligence analysis includes 21 algorithms and compares them against 30 benchmark tests. As a result, CSO and CHOA-B were identified as more universal, and 10 swarm intelligence algorithms were selected for comparison with TSOA. TSOA was compared with classical methods and five CEC benchmark suites, performing better on 45 out of 62 tests and 80% better than other swarm intelligence algorithms according to Wilcoxon's rank test. The TSOA leader selection algorithm was tested against GHS, Raft, and Random methods and was determined to be over 90% accurate, as well as time and memory efficient. Considering the quality of the results, instead of focusing solely on the cost of the solution, the leader selection algorithm should be made more efficient for use in large-scale applications. The proposed automatic fallen UAV capturing method is 89.47% faster and more resourceful. Simultaneously, the proposed task allocation method is also quite applicable for larger multi-UAV systems. This dissertation introduces a fully autonomous, distributed, time and memory-efficient leader selection algorithm, improving mission reliability and safety to support many large UAV applications.

CHAPTER I

INTRODUCTION

Unmanned Aerial Vehicle Systems (UAVS), also known as Drones, are aircraft operated without the presence of an onboard pilot. They can be either autonomously programmed to perform a particular set of tasks or controlled by a remote pilot. Nowadays, these UAVs are used for different purposes like agriculture [1], photography and videography [2], delivery [3], surveillance, military purposes [4], search and rescue [5], and many more. Multiple unmanned aerial vehicle systems (MUAVS) are groups of multiple UAVs that collaboratively intercommunicate and interact to perform a particular task or set of tasks [6]. A single remote pilot cannot control all those aircraft in multiple aerial vehicle systems; these systems use artificial intelligence to control the MUAVS.

Over the past decade, multiple unmanned aerial vehicle systems have gained significant traction, extending the capabilities of existing unmanned aerial vehicle applications. Researchers intended to focus on large UAV systems instead of single UAV systems. Military applications [7], coordinated drone delivery systems [8], emergency management [9], and drone shows [10] are the most commonly used applications in the world. Some applications involve more than 1000 individuals [10], and controlling them is more challenging. The hierarchical control system provides a more productive and scalable approach to managing large UAV swarms.

Nevertheless, local and global leadership significantly influences hierarchical control systems. In the absence of effective leadership, the entire group becomes vulnerable. Therefore, swiftly identifying this vulnerability and choosing appropriate leadership is essential [11]. Typical leader selection algorithms such as Bully, Paxos, Ring-based, Swarm intelligence-based, and GHS approaches are applicable to several distributed systems [12]-[18]. However, we identify four essential characteristics: autonomous, distributed, time-efficient, and resource constrained, which should describe a better leader selection algorithm. Current leader selection algorithms are developed for distributed large computing and ground multi-robot systems. Classical algorithms do not fulfill the aforementioned characteristics. This is a prominent research gap in the literature that this dissertation attempts to address.

Collective decision-making is one of the key processes in multi-robot systems. The best-of-n model describes which characteristics are essential for the collective decision-making process. This dissertation employs the best-of-n model to formulate the leader selection problem and its solution. Swarm intelligence imitates the collective behavior of nature and animals to solve various computational problems in distributed systems. This dissertation utilizes swarm intelligence to address the distributed leader selection problem.

The primary research objective of this dissertation is *to develop a novel, autonomous, and fully distributed leader selection algorithm to overcome the leader failure problem using a swarm intelligence approach, primarily focusing on enhancing time and memory efficiency in heterogeneous multiple autonomous unmanned aerial vehicles systems.*

This dissertation will propose an algorithm for leader selection in large heterogeneous multi-UAV systems. Shiyan and Anatoliy conceptualized hierarchical control for large multi-UAV systems [19]. Our approach can be used for multi-UAV military applications, drone light shows, and coordinated drone delivery systems that are hierarchically controlled. In addition, we propose a priority-based task allocation algorithm for drone swarms. We also provide an in-depth analysis of swarm intelligence algorithms. Then, we propose an algorithm based on tree trails called the "Tree Search Optimization Algorithm." Additionally, we introduce a method to capture a fallen agent in a UAV group.

This dissertation is structured as follows. Initially, CHAPTER II reviews the current state of the art in studies. This chapter describes best-of-n architectures, swarm intelligence algorithms, state-of-the-art multi-UAV systems, and current command structures. CHAPTER III covers the preliminaries of the model. CHAPTER IV provides the mathematical model of the system, current methods, and descriptions of the algorithm's models. Furthermore, this chapter details the experimental methods. CHAPTER V outlines the results. Finally, CHAPTER VI states the conclusion.

1.1 Motivation

UAV systems are more prevalent in many commercial and non-commercial applications. In recent years, multi-UAV applications have been the most popular in the military, entertainment, and research delivery. In May 2021, the Israel Defense Forces (IDF) used the Legion-X drone swarm in their military operations, designed and developed by Elbit Systems [20]. Legion-X is an autonomous heterogeneous robot swarm

engaged in networked combat operations, which enable many adversarial coordinated attacks on the battlefield [21]. The OFFensive Swarm-Enabled Tactics (OFFSET) program, launched by the Defense Advanced Research Projects Agency (DARPA), enabled a swarm of UAVs, including over 250 small unmanned aircraft systems (UASs), to perform given tasks in complex areas [22]. These swarm applications are autonomous and less efficient without ground controllers. Drone shows are also famous and widely used as massive 3D displays. According to Forbes, the DSF Drone Shows employ over 1000 small drones to build a 3D display in the sky with 16.7 million colors. Over 4000 individual drones were used in the Dubai World Cup horse racing event [10].

A centralized controlling approach offers many advantages to multi-robot systems, such as consistent decision-making, rigid control and coordination, effective resource allocation, stricter supervision, and simplicity. However, scalability is a limitation of centralized control. To overcome this issue, hierarchical control provides significant support. However, the malfunction of one agent or two combined agents increases the chance of failure for other follower agents. This is an essential problem for multi-UAV applications [23], [24]. Due to my research interest in artificial intelligence and intelligent robotics, I focus on developing a distributed, autonomous, lightweight, and efficient method to restabilize the control structure. I associate my problem-solving, programming, and software engineering skills with developing a framework for large UAV systems.

CHAPTER II

BACKGROUND

This chapter delves into the contextual foundational work pertinent to the present research. It presents the state of the art of leader election algorithms and studies of best-of-n modal. Further, it details a comprehensive analysis of 21 swarm intelligence algorithms and their core algorithm structure. Additionally, it outlines the state of the art of multi-UAV systems and leader election algorithms.

2.1 Best-of-n problem and its formalities

Collaborative decision-making is one of the essential tasks of swarm robotic applications. Swarm robots are extensive types of multi-robot systems. This chapter presented the different decision-making solutions under the influence of the work of Valentini et al. [25]. Collective decision-making is a well-formed activity of grouped animal organisms in nature. For example, the hunting behavior of grey wolves is one of the exciting behaviors observed in packed animals. By imitating these effective behaviors of nature, multi-robot systems, and swarm robot systems can take advantage of achieving complex tasks through teamwork.

Multi-robot systems and swarm robot systems have common characteristics. Multi-robot systems can be organized and controlled in a centralized or decentralized

manner. Instead, swarm robotics mainly acquires the advantage of a decentralized structure. Multi-robot systems have a limited number of robots, and swarm robot systems are equipped with many robots. Multi-robot and swarm robot systems are designed to achieve coordinated tasks and diverse applications. Swarm robot systems are robust and more flexible. Thus, swarm robot systems are a subgroup of multi-robot systems.

Collective decision-making is used in two different aspects of multi-robot systems: task allocation and consensus achievement problems. Task allocation is allocating the optimum task to the best robot agent. Consensus achievement is defined as collectively making the best decision [26]. Due to multi-robot systems being finite and countable, the consensus achievement problem is discrete.

2.1.1 Classification of different collective decision-making scenarios

The problem domain consists of the set of options O , and the multi-robot system should collectively decide which option is suitable out from n several options. This option set is a discrete set which can be defined by $O = \{O_1, O_2, \dots\}$. Here, it is assumed that every robot in the group makes its own decisions. the group of robots consisted with m number of robots and set of robots R can be defined as $R = \{R_1, R_2, \dots, R_m\}$. Majority of robots is defined by M , which is $M \leq (1 - \delta)N$ where $\delta, 0 \leq \delta \ll 0.5$, agreed with the option i , represented as the **collective decision**. When $\delta = 0$, all the robots are agreed to option i , the group of robots makes the consensus decision.

Each option has two characteristics: quality and cost. When robots make the consensus opinion O_i , that is described with the opinion quality ρ_i and the opinion quality σ_i . Both ρ_i and σ_i are normalized. The decision-making process aims to maximize the

opinion quality and minimize opinion cost. According to the relationship between opinion quality and cost, the above problems can be categorized into five categories. *Symmetric* means there is no difference between two options from the cost or quality point of view, and *Asymmetric* is vice versa. *Synergic* means no relationship exists between the option quality and cost, and *Antagonistic* is the reverse.

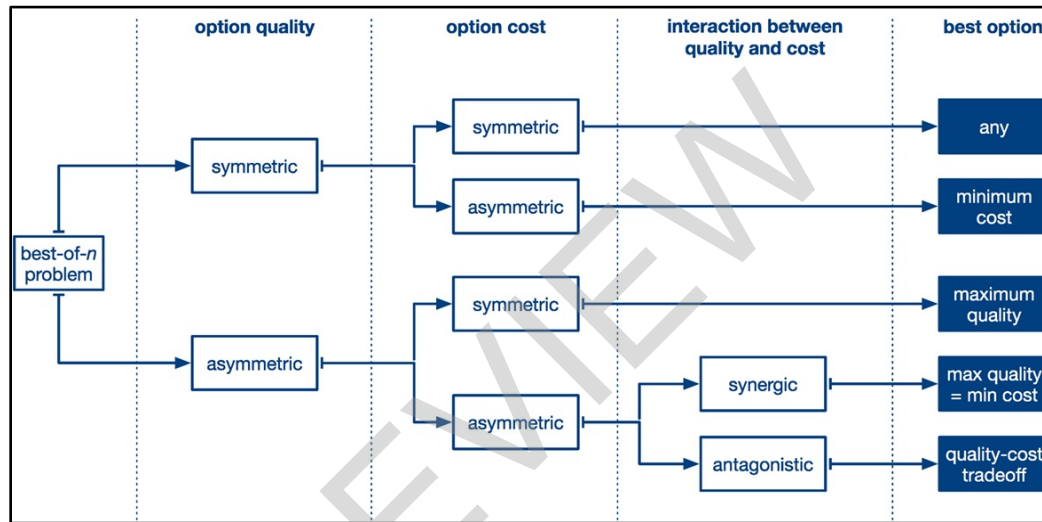


Figure 1. Different categories of the best-of-n problem, based on various combinations of option quality and option cost.

2.1.1.1 Symmetric Option Cost and Quality.

If any two options are selected from the options set with no differences in cost or quality, any option can be chosen as the best option. This scenario is called a Symmetry-breaking problem. Prey hunting scenario is one of the Symmetry-breaking problems. Robot swarms must capture prey from two places; first, hunted prey is the solution. Each prey value is the same, which means the opinion quality is symmetric. The prey are at the